

What Conformal Gravity Must Assume to Explain Galactic Rotation Curves

Claude Opus 5*

Asger Alstrup Palm†

Working draft, 6 August 2026

Abstract

Conformal (Weyl) gravity admits a static spherically symmetric vacuum solution whose metric function carries a term linear in r . That term is the basis of a long-standing proposal to fit galactic rotation curves without dark matter. We ask a reverse-physics question about it: not whether the fits succeed, but *what must be assumed for them to mean anything*.

Four results, each an exact symbolic computation with machine-checked controls. (i) The linear term is not an ansatz: in the conformal gauge the Bach vacuum equation is exactly the *biharmonic* equation $\nabla^4 B = 0$, so γr is the point-source response of the governing operator, on the same footing as the Newtonian $1/r$. (ii) If the baryonic Tully–Fisher relation holds, γ is forced to be a universal constant and is identified with the MOND acceleration scale, $\gamma = a_0/2c^2$; a γ carrying a mass-proportional piece gives the wrong slope. (iii) Conformal invariance forces the source to be trace-free, so a perfect fluid must satisfy $p = \rho/3$ and pressureless baryons are excluded outright. (iv) The standard repair—breaking the symmetry with a conformally coupled scalar—fails in its simplest form: a *constant* vacuum expectation value pins the Ricci scalar and thereby forces $\gamma = 0$, removing the very term at issue.

The conclusion is a trichotomy. Exactly one branch survives, and on it the question “is γ universal?” becomes “is the scalar profile universal across galaxies?”—a question about the matter sector rather than about gravity. We do not adjudicate it. We locate it.

1 The question

Conformal gravity has been proposed as an explanation of galactic rotation curves without dark matter, on the strength of a term linear in r in its static spherically symmetric vacuum solution. The literature on whether the fits succeed is substantial and we add nothing to it.

We ask instead the reverse-physics question: *what has to be true for the explanation to be an explanation?* A fit with a per-galaxy free parameter is a description; a fit with a parameter forced by the theory is a prediction. Which of these the proposal is depends on facts that can be settled by computation, and this paper settles four of them.

Every statement below is an exact symbolic identity over $\mathbb{Q}$, carried out with unspecified functions rather than sampled fixtures, with controls designed to fail if the claim were vacuous. Certificates and verifier commands are listed in §8.

*Anthropic model; computations and drafting.

†Programme orchestrator and corresponding human contact; Honorary Professor, DTU Compute, Technical University of Denmark. asger@area9.dk.

Conventions. We work in $D = 4$, signature $(-, +, +, +)$, with the static spherically symmetric ansatz

$$ds^2 = -B(r) dt^2 + \frac{dr^2}{B(r)} + r^2 d\Omega^2,$$

the conformal gauge $g_{tt}g_{rr} = -1$. §6 addresses what that gauge costs. B_{ab} denotes the Bach tensor.

2 The linear term is forced, and it is a point-source response

Theorem 2.1 (Completeness without an ansatz). *Let B be any C^4 function, nonvanishing on an interval. Then $B_{ab} = 0$ identically if and only if*

$$(rB)'''' = 0 \quad \text{and} \quad c_1^2 - 3c_0c_2 = 1,$$

where $B = c_0/r + c_1 + c_2r + c_3r^2$.

The proof is a decomposition. With $L := (rB)''''$ and N the nonlinear remainder, the three independent Bach rows satisfy, as exact identities in B and its derivatives,

$$B_{\theta\theta} = \frac{N}{24r^2}, \quad B_{tt} = \frac{B(N + 2r^3BL)}{24r^4}, \quad B_{rr} = \frac{-N + 2r^3BL}{24r^4B},$$

so all three lie in the span of $\{N, r^3BL\}$, and both generators are recoverable from the rows. Hence $B_{ab} = 0$ forces $L = 0$ and $N = 0$.

The essential point is that L is *linear*. Substituting $u = rB$ turns $L = 0$ into $u'''' = 0$, whose solution space is the cubics by four integrations—no ansatz. On that space N collapses to the constant $4(1 + 3c_0c_2 - c_1^2)$, leaving one algebraic condition.

Theorem 2.2 (The vacuum equation is biharmonic). *In spherical symmetry $\nabla^2 f = f'' + 2f'/r$, and for unspecified B*

$$\nabla^4 B = B'''' + \frac{4B'''}{r} = \frac{1}{r} (rB)''''.$$

So Theorem 2.1's linear equation is $\nabla^4 B = 0$: the Bach vacuum condition says B is *biharmonic*. The forced solution space $\{1/r, 1, r, r^2\}$ is the radial biharmonic kernel, and r^3 and r^{-2} —its nearest neighbours—are verified not to lie in it.

Corollary 2.3 (γr is a point-source response). *For ∇^2 the point-source potential is $1/r$ and its coefficient is the mass; this is Newton, and the origin of $\beta = GM/c^2$. For ∇^4 the corresponding object is r , since $\nabla^2 r = 2/r$ is harmonic away from the origin.*

This is a stronger statement than “ γr survives the equation”. It is the term the equation *produces* for a localised source, and it sits on the same footing as the Newtonian term rather than being appended to it.

Remark 2.4. That the general static spherically symmetric vacuum solution of conformal gravity has this form is classical. What is contributed here is the removal of the ansatz and the identification of the operator, both computed for an unspecified function.

3 Tully–Fisher forces γ to be universal

With $\beta = GM/c^2$ the locally measured circular velocity is $v^2/c^2 = rB'/2B$, which at small field strength is $\beta/r + \gamma r/2 - kr^2$. In the galactic regime this has a stationary point, and it is a minimum:

$$r_* = \sqrt{2\beta/\gamma}, \quad v^4 = 2GM\gamma c^2 \left(1 - \frac{3}{2}\beta\gamma\right).$$

Theorem 3.1 (The conditional). *To leading order in $\beta\gamma$: if γ is universal then $v^4 \propto M$, the baryonic Tully–Fisher relation with slope 4 in v . If γ carries a piece proportional to M then $v^4 \propto M^2$. Matching the observed $v^4 = GMa_0$ forces*

$$\gamma = \frac{a_0}{2c^2}.$$

Read in the reverse-physics direction: the observed Tully–Fisher relation, combined with the forced metric, requires γ to be a universal constant of nature rather than a per-galaxy parameter, and identifies it with the MOND acceleration scale. That is a prediction rather than a fit, and it is falsifiable in a specific way—a mass-dependent γ moves the slope.

Two boundaries travel with it. Identifying $v_{\min}$ with the observed flat velocity is an interpretive step, not a theorem. And the correction factor $1 - \frac{3}{2}\beta\gamma$ is itself mass-dependent, so the slope is exactly 4 only in the limit $\beta\gamma \rightarrow 0$; for a galaxy the departure is of order 10^{-14} and unobservable, but “exactly linear” would be too strong.

Remark 3.2. Conformal gravity and MOND are not the same theory and disagree away from r_* : beyond it the conformal curve *rises* ($dv^2/dr|_{2r_*} = 3\gamma/8 > 0$) where MOND’s are asymptotically flat. Only the $-kr^2$ term can bend it back. The outer rotation curve is therefore a discriminant, not a detail.

4 What may source the theory

§2 and §3 concern the vacuum. γ is a free coefficient there; the explanatory claim needs it *sourced*.

A conformally invariant action satisfies $g_{ab} \delta S / \delta g_{ab} = 0$. Applied to the gravitational action this is the Noether identity of the theory; applied to the matter action it reads $T^\mu{}_\mu = 0$. It is one theorem used twice, and we take it as given.

Theorem 4.1 (Trace obstruction). *A perfect fluid has $T^\mu{}_\mu = 3p - \rho$, so it is trace-free if and only if $p = \rho/3$. Dust, $p = 0$, forces $\rho = 0$.*

Radiation, uniquely, and pressureless matter not at all. A galaxy’s baryons are pressureless to excellent approximation, so on the conformally invariant branch a galaxy is not an admissible source.

The standard repair is to break the symmetry with a conformally coupled scalar acquiring a vacuum expectation value, generating a scale so that ordinary matter can source the theory after all. In its simplest form this fails, and it fails on the term at issue.

Theorem 4.2 (A constant vacuum expectation value forces $\gamma = 0$). *For the conformally coupled scalar, $\square S + \frac{1}{6}RS + 4\lambda S^3 = 0$ reduces for constant $S = S_0 \neq 0$ to $R = -24\lambda S_0^2$, a constant. For the forced family,*

$$R = 12k - \frac{6\gamma}{r} + \frac{2(1-w)}{r^2},$$

which is constant in r if and only if $\gamma = 0$ and $w = 1$.

A constant vacuum expectation value does not merely supply a scale; it pins the Ricci scalar, and the forced family is compatible with a pinned R only where the linear term vanishes. Note that u , the Newtonian coefficient, drops out of R entirely: the obstruction is specific to γ .

5 The trichotomy

Collecting §4, the possibilities are organised by one question—*is there a scale, and is it constant or field-dependent?*

	scale	consequence
(A) conformal matter	none	source must be trace-free, hence radiation; a galaxy is excluded, so γ is not tied to M and Tully–Fisher is a coincidence
(B1) constant $\langle S \rangle$	constant	R pinned, hence $\gamma = 0$; the linear term is removed
(B2) non-constant S	field-dependent	γ tied to the scalar profile; open

(A) does not remove γ ; it decouples γ from mass, which destroys the explanation while leaving the term. (B1) removes γ outright. Exactly one branch survives, and it is the one the literature in fact uses.

Corollary 5.1 (Where the assumption lives). *On branch (B2), γ is determined by the scalar profile, which is a property of each configuration. Theorem 3.1 requires γ to be the same constant for every galaxy. Therefore the requirement transfers: is the scalar profile universal across galaxies?*

That is a question about the symmetry-breaking sector rather than about gravity, and we know of no reason internal to the gravitational theory for it to answer yes. This is the assumption a conformal-gravity rotation-curve fit is standing on.

6 What the conformal gauge costs

Sections 2–5 work in the gauge $g_{tt}g_{rr} = -1$. Two things are needed to justify it.

Proposition 6.1 (Conformal covariance). *In $D = 4$, $B_{ab}[\Omega^2 g] = \Omega^{-2} B_{ab}[g]$, verified for the Mannheim–Kazanas family with symbolic parameters and unspecified Ω . Hence Bach flatness is a property of the conformal class, and the classification of §2 classifies classes.*

Proposition 6.2 (Reachability is a first-order ODE). *Setting the new radius to $\rho = \Omega r$ so the angular part remains ρ^2 , the gauge condition becomes*

$$r \Omega' = \Omega(\Omega\sqrt{ab} - 1),$$

first order in Ω , hence locally solvable wherever $ab > 0$ and $r \neq 0$.

What remains assumed is only global continuation—whether Ω stays positive and finite across the exterior. That is strictly smaller than the original assumption and is a question of analysis rather than algebra.

7 What this paper does not establish

- **No novelty in the physics.** The Mannheim–Kazanas solution, its reduction to a fourth-order Poisson problem, and the trace-free condition on conformal matter are all classical. The contribution is that each is computed here with the ansatz removed and the assumptions isolated, rather than cited.
- **No fits.** No galaxy data enters anywhere. The only observational inputs are the *shape* of the Tully–Fisher relation, used as a premise in Theorem 3.1, and that galaxies are pressureless, used in §4. Neither is our measurement.
- **No verdict on branch (B2).** A non-constant scalar is not analysed. Theorem 4.2 shows the non-constancy is load-bearing rather than incidental; it does not show the branch fails.
- **No resolution of the Flanagan–Mannheim dispute** [3, 4] over the Newtonian limit under conformal matter coupling. Both positions choose branches of §5, and on either side the choice is an assumption. We locate the disagreement; we do not settle it.
- **Nothing about the ghost.** Conformal gravity’s fourth-order kinetic operator carries an Ostrogradsky ghost. That is orthogonal to everything here and is treated elsewhere in this programme.
- **No numerical value.** a_0 is not measured here and no published γ is compared; $\gamma = a_0/2c^2$ is an algebraic consequence of a premise, not a numerical agreement.

8 Verification

Each result is a fail-closed certificate with a deterministic verifier. Controls are designed to fail if the claim were vacuous; several did so during development and are recorded in the certificates.

result	certificate	checks
Thm. 2.1	BHOB_GENERAL_STATIC_SPHERICAL_COMPLETENESS	24
Thm. 3.1	BHOC_TULLY_FISHER_SCALING	14
Thm. 2.2	BHOD_BACH_VACUUM_IS_BIHARMONIC	9
Thm. 4.1	BHOE_MATTER_TRACE_DILEMMA	8
Thm. 4.2	BHOF_CONSTANT_VEV_FORCES_GAMMA_ZERO	12
§6	BHOG_CONFORMAL_CLASS_INVARIANCE	10

Theorems 2.1 and 4.2 and §6 are computed in SYMPY; Theorems 2.2 and 4.1 are computed in Forge over exact rationals with symbolic differentiation of unspecified functions. Theorem 2.1’s conclusion carries an independent Forge rail using a different arithmetic stack, representation and code path; that rail confirms which metrics are Bach-flat but not the row decomposition itself, which remains single-implementation.

References

- [1] P. D. Mannheim and D. Kazanas, *Exact vacuum solution to conformal Weyl gravity and galactic rotation curves*, *Astrophys. J.* **342** (1989) 635.
- [2] R. J. Riegert, *Birkhoff’s theorem in conformal gravity*, *Phys. Rev. Lett.* **53** (1984) 315.

- [3] É. É. Flanagan, *Fourth order Weyl gravity*, Phys. Rev. D **74** (2006) 023002.
- [4] P. D. Mannheim, *Alternatives to dark matter and dark energy*, Prog. Part. Nucl. Phys. **56** (2006) 340.
- [5] M. Milgrom, *A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis*, Astrophys. J. **270** (1983) 365.